Document register

SO3 Converter Ver 1 · SO3-V1-PM-DOC-001 · Rev 0

Department of Chemical Engineering, Chulalongkorn University (teaching)

REV	DATE	DESCRIPTION	ORIGINATOR	CHECKED / APPROVED
0	27 Oct 2026	Issued for teaching	Apinan Soottitantawat	—

Generated for teaching from the SO3 Converter Ver 1 operator training simulator. Every number comes from the simulator model (design point, noise off) or is marked assumed. Not for construction. No client, contractor or vendor is represented.

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1 · Purpose and scope

This register lists every document issued for the converter section of SO3 Converter Ver 1 (the operator training simulator), shows how the documents feed each other, and collects the engineering queries raised while writing them. It is generated by tools/gen.js from the PDF files present (number, title, revision, pages, date and the document numbers each PDF cites are read from the files) and from the tabs that hold the two documents with no PDF. The instructor workbook is a training aid, not a project document, and is not registered.

2 · How an EPC project and its documents flow (typical practice)

In an EPC (engineering, procurement, construction) contract one contractor designs the plant, buys the equipment, builds it and hands it over running. FEED fixes the process, the PFD, the heat and material balance and the main equipment; detailed engineering turns them into P&IDs, datasheets, lists and studies (HAZOP, relief, SIL); procurement buys against the datasheets; construction builds to the issued-for-construction drawings; commissioning proves the plant and hands over the as-built set.

Phase	Main output	Revision status the documents typically carry
FEED	PFD, HMB, main equipment, plot plan	IFR → IFA
Detailed engineering	P&IDs, datasheets, lists, HAZOP, relief, SIL, C&E, load list	IFR → IFA → IFC
Procurement	requisitions, orders, vendor data, FAT	IFP, certified vendor data
Construction	the plant, built to IFC; redlines	IFC + redlines
Commissioning	MC, pre-commissioning, start-up, performance test, handover dossier	As-built

2.1 Revision codes

Code	Meaning	Revision mark	What it allows
IFR	Issued for review	A, B, C …	Other disciplines and the owner comment; the document may still change a lot.
IFA	Issued for approval	next letter	Comments are in; the owner approves the basis.
IFC	Issued for construction	0, 1, 2 …	Frozen for purchase and build; every later change is a numbered revision with a clouded change.
As-built	As built	next number	Construction redlines incorporated; the record the operators keep.
IFT	Issued for teaching (this set)	0 (new set); A (P&IDs, HAZOP)	Generated from the simulator model; nobody checked or approved it (EQ-ej-03).

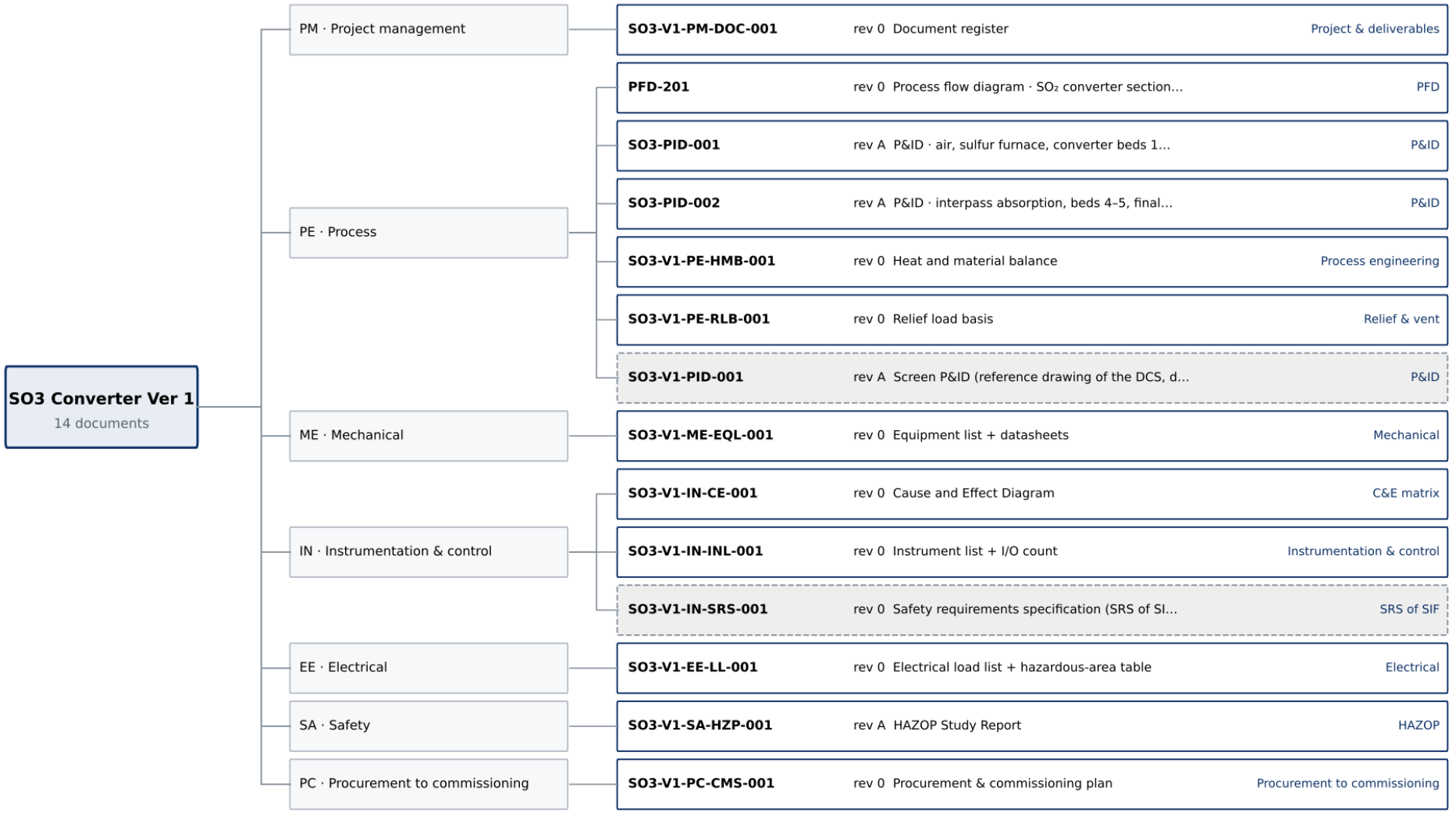
2.2 Discipline codes and numbering scheme

Every new document is numbered SO3-V1---: project SO3, unit V1, the discipline code, a document-type code and a three-digit sequence.

Code	Discipline	Documents in this register
PM	Project management	SO3-V1-PM-DOC-001 (DOC)
PE	Process	PFD-201 (PFD), SO3-PID-001 (P&ID 1), SO3-PID-002 (P&ID 2), SO3-V1-PE-HMB-001 (HMB), SO3-V1-PE-RLB-001 (RLB), SO3-V1-PID-001 (Screen P&ID)
ME	Mechanical	SO3-V1-ME-EQL-001 (EQL)
IN	Instrumentation & control	SO3-V1-IN-CE-001 (CE), SO3-V1-IN-INL-001 (INL), SO3-V1-IN-SRS-001 (SRS)
EE	Electrical	SO3-V1-EE-LL-001 (LL)
SA	Safety	SO3-V1-SA-HZP-001 (HZP)
PC	Procurement, construction & commissioning	SO3-V1-PC-CMS-001 (CMS)

3 · Deliverable tree by discipline

Every document by discipline, with the tab of the OTS page that holds it. Grey dashed boxes are held only in a tab (no PDF issued).



4 · Document register

Status of every document: Issued for teaching. File names are relative to the published page (all files sit beside it).

Doc no.	Title	Disc.	Type	Rev	Status	Originator	Pages	Date	Tab	File
SO3-V1-PM-DOC-001	Document register	PM	DOC	0	Issued for teaching	Apinan Soottitantawat	13	2026-10-27	Project & deliverables	SO3-V1-PM-DOC-001.pdf
PFD-201 (≠ scheme)	Process flow diagram · SO_2 converter section, sulfur-burning H_2SO_4 plant, 125 t/h	PE	PFD	0	Issued for teaching	Apinan Soottitantawat	1	2026-10-19	PFD	pfd-201.pdf
SO3-PID-001 (≠ scheme)	P&ID · air, sulfur furnace, converter beds 1–3	PE	PID	A	Issued for teaching	Apinan Soottitantawat	1	25-SEP-26	P&ID	SO3-PID-001.pdf
SO3-PID-002 (≠ scheme)	P&ID · interpass absorption, beds 4–5, final absorption	PE	PID	A	Issued for teaching	Apinan Soottitantawat	1	25-SEP-26	P&ID	SO3-PID-002.pdf
SO3-V1-PE-HMB-001	Heat and material balance	PE	HMB	0	Issued for teaching	Apinan Soottitantawat	16	2026-10-27	Process engineering	SO3-V1-PE-HMB-001.pdf
SO3-V1-PE-RLB-001	Relief load basis	PE	RLB	0	Issued for teaching	Apinan Soottitantawat	9	2026-10-27	Relief & vent	SO3-V1-PE-RLB-001.pdf
SO3-V1-PID-001 (≠ scheme)	Screen P&ID (reference drawing of the DCS, decks and exercises)	PE	PID	A	Issued for teaching	—	—	—	P&ID	— (held in the tab)
SO3-V1-ME-EQL-001	Equipment list + datasheets	ME	EQL	0	Issued for teaching	Apinan Soottitantawat	13	2026-10-27	Mechanical	SO3-V1-ME-EQL-001.pdf
SO3-V1-IN-CE-001	Cause and Effect Diagram	IN	CE	0	Issued for teaching	Apinan Soottitantawat	2	2026-10-19	C&E matrix	SO3-V1-IN-CE-001.pdf
SO3-V1-IN-INL-001	Instrument list + I/O count	IN	INL	0	Issued for teaching	Apinan Soottitantawat	10	2026-10-27	Instrumentation & control	SO3-V1-IN-INL-001.pdf
SO3-V1-IN-SRS-001	Safety requirements specification (SRS of SIF)	IN	SRS	0	Issued for teaching	Apinan Soottitantawat	—	—	SRS of SIF	— (held in the tab)
SO3-V1-EE-LL-001	Electrical load list + hazardous-area table	EE	LL	0	Issued for teaching	Apinan Soottitantawat	7	2026-10-27	Electrical	SO3-V1-EE-LL-001.pdf
SO3-V1-SA-HZP-001	HAZOP Study Report	SA	HZP	A	Issued for teaching	HAZOP team (teaching study)	81	2026-10-19	HAZOP	SO3-V1-SA-HZP-001.pdf
SO3-V1-PC-CMS-001	Procurement & commissioning plan	PC	CMS	0	Issued for teaching	Apinan Soottitantawat	14	2026-10-27	Procurement to commissioning	SO3-V1-PC-CMS-001.pdf

4.1 Worked calculation · engineering progress by rules of credit

Rules of credit (typical practice, percentages assumed): Not started 0 %, Started 10 %, IFR 40 %, IFA 70 %, IFC 100 %, As-built 100 %. Weight = PDF page count (a proxy for effort, assumed); a tab-only document counts 2.

Example (assumed status): FEED documents frozen, lists and studies issued for approval, the rest for review — IFC: PFD, P&ID 1, P&ID 2, HMB (Σw = 19); IFA: EQL, CE, INL, SRS, HZP (Σw = 108); IFR: DOC, RLB, Screen P&ID, LL, CMS (Σw = 45).

Progress = Σ(w·credit)/Σw = (19 × 100 % + 108 × 70 % + 45 × 40 %) / 172 = 112.6 / 172 = 65.5 %.

The HAZOP report alone carries 81 of the 172 pages: page count is a crude weight; real projects weight by budgeted man-hours. The tab has a calculator that also flags a document issued ahead of its inputs.

5 · Interface matrix

Row = the document that gives information; column = the document that uses it. The interfaces are the typical EPC flow (design intent); the mark says whether the receiving PDF cites the input by number (read from its text): ● cited, ○ expected but not cited, ◆ receiving document has no PDF.

feeds → uses	DOC	PFD	P&ID 1	P&ID 2	HMB	RLB	Screen P&ID	EQL	CE	INL	SRS	LL	HZP	CMS
DOC														
PFD			○	○	●			●						
P&ID 1						○		●	○	●			●	●
P&ID 2						○		●	○	●			●	●
HMB						○		●		○		○		●
RLB														
Screen P&ID										○			●	
EQL						●						●		●
CE										●				●
INL												○		●
SRS									●	●				●
LL														●
HZP						○			○		◆			
CMS														

5.1 Interfaces and what passes

From	To	What passes	Cited in the receiving PDF
PFD-201	SO3-PID-001	equipment, streams, control scheme	NO
PFD-201	SO3-PID-002	equipment, streams, control scheme	NO
PFD-201	SO3-V1-PE-HMB-001	stream numbers and equipment	yes
PFD-201	SO3-V1-ME-EQL-001	equipment items and tags	yes
SO3-V1-PE-HMB-001	SO3-V1-ME-EQL-001	duties, flows, operating pressure and temperature	yes
SO3-V1-PE-HMB-001	SO3-V1-PE-RLB-001	flows and conditions for relief loads	NO
SO3-V1-PE-HMB-001	SO3-V1-IN-INL-001	normal values for ranges	NO
SO3-V1-PE-HMB-001	SO3-V1-EE-LL-001	blower and pump duties	NO
SO3-V1-PE-HMB-001	SO3-V1-PC-CMS-001	performance-test targets	yes
SO3-PID-001	SO3-V1-IN-INL-001	instrument tags and loops	yes
SO3-PID-002	SO3-V1-IN-INL-001	instrument tags and loops	yes
SO3-PID-001	SO3-V1-PE-RLB-001	relief devices, isolation, line routing	NO
SO3-PID-002	SO3-V1-PE-RLB-001	relief devices, isolation, line routing	NO
SO3-PID-001	SO3-V1-IN-CE-001	interlock schedule	NO
SO3-PID-002	SO3-V1-IN-CE-001	interlock schedule	NO
SO3-PID-001	SO3-V1-SA-HZP-001	study drawing (nodes)	yes
SO3-PID-002	SO3-V1-SA-HZP-001	study drawing (nodes)	yes
SO3-PID-001	SO3-V1-ME-EQL-001	equipment, nozzles, line numbers	yes
SO3-PID-002	SO3-V1-ME-EQL-001	equipment, nozzles, line numbers	yes
SO3-PID-001	SO3-V1-PC-CMS-001	systems for MC and pre-commissioning	yes

From	To	What passes	Cited in the receiving PDF
SO3-PID-002	SO3-V1-PC-CMS-001	systems for MC and pre-commissioning	yes
SO3-V1-PID-001	SO3-V1-SA-HZP-001	study drawing used by the team	yes
SO3-V1-PID-001	SO3-V1-IN-INL-001	DCS tags and alarms	NO
SO3-V1-SA-HZP-001	SO3-V1-PE-RLB-001	relief scenarios	NO
SO3-V1-SA-HZP-001	SO3-V1-IN-SRS-001	hazards and demands (via LOPA)	no PDF
SO3-V1-SA-HZP-001	SO3-V1-IN-CE-001	trips and alarms recommended	NO
SO3-V1-IN-SRS-001	SO3-V1-IN-CE-001	SIF settings, delays, safe states	yes
SO3-V1-IN-SRS-001	SO3-V1-IN-INL-001	SIS tags and architecture	yes
SO3-V1-IN-SRS-001	SO3-V1-PC-CMS-001	SAT and proof tests	yes
SO3-V1-IN-CE-001	SO3-V1-IN-INL-001	trip and alarm settings	yes
SO3-V1-IN-CE-001	SO3-V1-PC-CMS-001	SAT steps	yes
SO3-V1-ME-EQL-001	SO3-V1-PE-RLB-001	design pressures and temperatures	yes
SO3-V1-ME-EQL-001	SO3-V1-EE-LL-001	drives and heaters	yes
SO3-V1-ME-EQL-001	SO3-V1-PC-CMS-001	procurement packages	yes
SO3-V1-IN-INL-001	SO3-V1-EE-LL-001	instrument and UPS loads	NO
SO3-V1-IN-INL-001	SO3-V1-PC-CMS-001	loop checks, FAT and SAT	yes
SO3-V1-EE-LL-001	SO3-V1-PC-CMS-001	energisation, electrical packages	yes

6 · Engineering-query register

58 queries, all OPEN: the queries.json of each Engineering tab, the P&ID tab's recorded differences (part 5), the open items of the project resume and this register's own findings. Sources: Procurement to commissioning 5, Electrical 4, Instrumentation & control 9, Mechanical 6, Process engineering 8, Relief & vent 8, P&ID tab 8, Project resume 5, Project & deliverables 5. Queries that say the same thing are grouped under 11 linked keys (30 queries), each keeping its original id. Tracked in their own tabs and not repeated: 43 HAZOP recommendations and 26 recommended additions of the 104-item safety-layer register.

6.1 Linked keys

Key	Subject	Queries (original ids)	Raised in
L-01	B-101 duty: 61.6 MW (model, gas side) against ≈ 45 MW (44.8 MW boiling only)	EQ-ep-01, EQ-em-02, EQ-rs-01	Mechanical, Process engineering, Project resume
L-02	K-101 shaft power 4200 kW implies an isentropic efficiency of 0.95–1.06	EQ-ep-03, EQ-em-04, EQ-ee-01	Electrical, Mechanical, Process engineering
L-03	Single sulfur shut-off XV-103: SIS architecture does not reach the required SIL	EQ-ei-08, EQ-ec-02	Procurement to commissioning, Instrumentation & control
L-04	SIS trips share their sensors with the DCS indication and alarm	EQ-ei-01, EQ-ei-05, EQ-pd-04	Instrumentation & control, P&ID tab
L-05	Pass-1 / bed-1 design temperature against the trip and the excursions the model shows	EQ-ep-06, EQ-em-01, EQ-er-04	Mechanical, Process engineering, Relief & vent
L-06	Steam system (drum, BFW, blowdown) outside the model and not documented	EQ-ep-07, EQ-er-08, EQ-ec-03	Procurement to commissioning, Process engineering, Relief & vent
L-07	Rate push to 45 t/h: PI-200 above its 42 kPa g HI alarm	EQ-ep-04, EQ-er-07	Process engineering, Relief & vent
L-08	P&ID rev B: design-point table and missing instruments	EQ-em-03, EQ-ei-06, EQ-ei-07, EQ-pd-05, EQ-pd-06	Instrumentation & control, Mechanical, P&ID tab
L-09	Heat of sulfur combustion 295 against 296.8 kJ/mol	EQ-ep-02, EQ-rs-02	Process engineering, Project resume
L-10	Drawing numbers outside the numbering scheme	EQ-pd-01, EQ-ej-01	P&ID tab, Project & deliverables
L-11	Gas-train design pressure basis from the K-101 shut-off	EQ-er-01, EQ-em-05	Mechanical, Relief & vent

6.2 Register (grouped by linked key, then the unlinked queries)

Key	EQ id	Document	Discipline	Subject	Finding	Proposed resolution	Raised by	Status
L-01	EQ-em-02	SO3-V1-SA-HZP-001	Process / Mechanical	B-101 duty quoted as ≈ 45 MW	HAZOP node 3 intent and the safety-layer register give B-101 ≈ 45 MW; the model gives 61.6 MW (94.0 t/h × (2801 − 440) kJ/kg), used on the datasheet.	Correct the HAZOP and register text to 61.6 MW; base the PSV-101 relief load (Relief & vent tab) on 94 t/h.	Mechanical	OPEN
L-01	EQ-ep-01	SO3-V1-PE-HMB-001	Process	B-101 duty basis	The HMB gas-side duty of B-101 is 61.6 MW (= FI-108 × 2361 kJ/kg, the model raises steam from water at 440 kJ/kg). The Process tab quotes 44.8 MW (= FI-108 × (2801 − 1087) kJ/kg, boiling only). The BFW is already heated in E-203/E-204/E-205 (49.5 MW), so if the economizer water feeds the B-101 drum, the model counts the BFW preheat twice.	Use the gas-side duty 61.6 MW for the B-101 datasheet; state the steam-side split (sensible + boiling) on the steam-system HMB and align the Process tab text.	Process engineering	OPEN
L-01	EQ-rs-01	SO3-V1-PE-HMB-001 / tabs/BRIEF.md	Process	B-101 duty in the brief	The agent brief and the Process tab quote B-101 ≈ 45 MW (44.8 MW boiling only); the model raises 94 t/h steam from 440 kJ/kg water, 61.6 MW on the gas side.	Quote 61.6 MW (gas side) everywhere and give the sensible/boiling split on the steam-system HMB; correct the brief.	Project resume	OPEN
L-02	EQ-ee-01	SO3-V1-EE-LL-001	Electrical / Mechanical	K-101 motor rating basis	The load list uses the model blower power 4200 kW (absorbed). The Mechanical hand check (EQ-em-04) implies η 0.95; at η 0.80 the shaft power would be 4978 kW. The 5000 kW motor covers it with 0.5 % margin only.	Hold the motor rating at 5000 kW until the vendor curve is in; confirm before ordering the motor and the 6.6 kV feeder.	Electrical	OPEN
L-02	EQ-em-04	SO3-V1-ME-EQL-001	Mechanical / Electrical	K-101 shaft power vs isentropic hand check	Model shaft power 4200 kW; isentropic power from 45 °C, −1 kPa g suction (assumed) to 60 kPa g is 3982 kW → implied isentropic efficiency 0.95 (typical 0.80–0.85, i.e. 4684–4977 kW). The model discharge 90 °C equals the isentropic outlet 91 °C.	Confirm suction temperature and efficiency with the blower vendor; size the motor on the vendor curve, not on the model power (see EQ-ee-01).	Mechanical	OPEN
Key	**EQ id**	**Document**	**Discipline**	**Subject**	**Finding**	**Proposed resolution**	**Raised by**	**Status**
L-02	EQ-ep-03	SO3-V1-PE-HMB-001	Process / Mechanical	K-101 power vs gas power	With the model's suction at 90.0 °C and atmospheric pressure, the isentropic gas power for 238.5 kNm³/h and 60.0 kPa is 4440 kW, above the 4200 kW shaft power in econ() (implied efficiency 106 %). The model	Confirm whether 90 °C is the blower suction or discharge temperature. With 45 °C suction (typical after the drying tower, assumed) the isentropic power is 3889 kW; at	Process engineering	OPEN

Key	EQ id	Document	Discipline	Subject	Finding	Proposed resolution	Raised by	Status
					also gives no temperature rise across K-101 (stream 2 = stream 1 = 90.0 °C); 4200 kW into the air would raise it ≈ 48.8 K.	80–85 % efficiency the shaft power is ≈ 4576–4862 kW. Size the motor on that (electrical load list).		
L-03	EQ-ec-02	SO3-V1-PC-CMS-001 / SO3-V1-IN-SRS-001	Instrumentation (SIS)	SIS architecture not fixed before the SIS purchase order	The SRS required SIL is not met by the 1oo1 architecture for 5 SIF(s): SIF-01 FSLL-100: required SIL 2 (PFD ≤ 1.6e-3), achieved SIL 1 (PFD 1.8e-2); SIF-02 TSHH-203: required SIL 2 (PFD ≤ 8.9e-3), achieved SIL 1 (PFD 1.3e-2); SIF-03 FSLL-301: required SIL 2 (PFD ≤ 3.3e-3), achieved SIL 1 (PFD 1.8e-2); SIF-04 FSLL-302: required SIL 3 (PFD ≤ 4.8e-4), achieved SIL 1 (PFD 1.8e-2); SIF-05 ASHH-306: required SIL 1 (PFD ≤ 1.7e-2), achieved SIL 1 (PFD 3.8e-2). PK-11 and the XV-103 valve in PK-10 cannot be bought or FAT-tested against a final C&E until this is closed.	Decide the architecture (e.g. a second shut-off valve XV-103B in 1oo2, redundant transmitters) before the PK-11 MR; re-verify the PFD; revise the C&E and the SAT steps.	Procurement to commissioning	OPEN
L-03	EQ-ei-08	SO3-V1-IN-INL-001	Instrumentation / SIS	Second sulfur shut-off goes through the DCS	Every trip closes XV-103 (SIS) and forces FIC-103 to MAN 0 % so FV-103 also closes, but that second action is a DCS command over the gateway, not an SIS output.	Add an SIS solenoid XY-103B on FV-103 (Safety layers part 5): +1 SIS DO, inside the 20 % spare.	Instrumentation & control	OPEN
L-04	EQ-ei-01	SO3-V1-IN-INL-001	Instrumentation	SIF sensors shared with the DCS (FT-100, TE-203, FT-301, FT-302, AIT-306)	Each of the five SIS trips (FSLL-100, TSHH-203, FSLL-301, FSLL-302, ASHH-306) reads the same transmitter as the DCS indication and alarm; FT-301 also drives the P-301B auto-start and FT-100 the H-101 low-air interlock. One failed transmitter (frozen or reading on the safe side) defeats the alarm and the trip together. The instrument list carries them as SIS AI through a signal splitter (assumed).	Provide dedicated SIS transmitters (TE-203B, FT-301B, FT-100B, FT-302B; ASHH-306 treated as an environmental trip or given a second analyzer), wired to SIS-100, with the DCS reading a repeat. Spares in the SIS AI count cover one; the rest add SIS AI. Cross-reference: Safety layers part 5, SRS part 5, HAZOP recommendations.	Instrumentation & control	OPEN
L-04	EQ-ei-05	SO3-V1-IN-INL-001	Instrumentation	FSLL-100 trip low in the FT-100 orifice range	FSLL-100 trips at 90 kNm³/h, 30 % of the 0–300 kNm³/h range of FT-100: an orifice gives only 9 % of its differential-pressure span there ($\Delta P \propto Q^2$), where accuracy is poor (typical orifice turndown 3–4 : 1).	Dedicated SIS flow transmitter (see EQ-ei-01) ranged for the trip (e.g. 0–150 kNm³/h), or a low-range DP cell; confirm with the flow element calculation.	Instrumentation & control	OPEN
L-04	EQ-pd-04	SO3-PID-001/002 rev A / SO3-V1-PID-001 rev A	Instrumentation / SIS	SIS functions drawn as separate bubbles	The EPC sheets show FSLL-100, TSHH-203, FSLL-301, FSLL-302 and ASHH-306 as SIS functions (circle in a square). In the model each trip reads the same measurement as the DCS indication and alarm (FI-100, TI-203, FI-301, FI-302, AIT-306), so the sheets' bubbles share their sensors with the DCS. The reference drawing has no separate trip bubbles: it shows the SIS through XV-103 (S actuator), AIT-306 in a square and its note 2.	Separate SIS transmitters for TSHH-203, FSLL-100, FSLL-301/302 and ASHH-306 (P&ID tab part 6, HOLD).	P&ID tab	OPEN
L-05	EQ-em-01	SO3-V1-ME-EQL-001	Mechanical / Safety	R-201 pass 1, SH-201, E-201 design temperature vs SIF-02 response	Pass-1 design temperature is 700 °C (TSHH-203 660 °C + margin; the SRS also assumes 700 °C). The SRS process-safety-time run of SIF-02 shows the bed 1 exit still peaking at 744.1 °C with the SIF acting (791.8 °C without).	Either raise the pass-1 design temperature to 750 °C (thicker 304H shell, see EQ-em-05) or add the bed-1 inlet trip TSHH-201B recommended in the SRS so the exit stays near 631 °C; decide in the SRS/LOPA review.	Mechanical	OPEN
L-05	EQ-ep-06	SO3-V1-PE-HMB-001	Process / Mechanical	Bed 1 exit design temperature	The typical rule (operating + 30 °C) on the highest bed 1 exit of the three cases (632.3 °C) gives 665 °C, only 5 K above the TSHH-203 trip at 660 °C; a runaway reaches the trip before the SIS acts (5 s delay) and can overshoot.	Set the R-201 pass-1 shell / internals design temperature on the trip setting plus overshoot (e.g. ≥ 680 °C, to be confirmed by the Mechanical tab and the catalyst vendor limit).	Process engineering	OPEN
L-05	EQ-er-04	SO3-V1-PE-RLB-001	Mechanical	TV-201 failing open: temperature basis of the bed-1 inlet duct	TV-201 wide open drives TI-201 to 718 °C (model, trips off) against an R-201 / bed-1 inlet design temperature of 700 °C.	Design the B-101 outlet/bed-1 inlet duct and the R-201 top shell (or their refractory) for the TV-201 open case, or limit TV-201 travel with a mechanical stop sized from the model.	Relief & vent	OPEN
L-06	EQ-ec-03	SO3-V1-PC-CMS-001	Mechanical / Process	Steam drum and BFW system outside the model	PK-03 needs the steam drum, blowdown and BFW pumps to raise 94 t/h at 4.0 MPa g, but the model and PFD-201 stop at B-101 / the economisers (BFW 150 t/h). Boil-out, steam blowing and the performance test of steam export depend on them.	Define the steam-system battery limit; add the drum and BFW equipment to the EQL and to PK-03 or a utility package; include them in the WHB & steam system of the MC checklist.	Procurement to commissioning	OPEN
L-06	EQ-ep-07	SO3-V1-PE-HMB-001	Process	Steam / BFW balance	BFW to the economizers FI-253 = 150 t/h but steam FI-108 = 93.99 t/h; the model treats the economizer water and the boiler as separate circuits, so the steam-drum balance (blowdown, deaerator, export) cannot be closed from the model.	Issue a utility HMB for the steam system (drum, blowdown ≈ 1–2 % typical, deaerator) with the economizer water routed to the drum.	Process engineering	OPEN
L-06	EQ-er-08	SO3-V1-PE-RLB-001	Process	Steam side is outside the model	Steam loads come from the model gas side (FI-108 94.0 t/h design, 103.8 t/h at 45 t/h sulfur, 118.0 t/h with TV-201 closed); drum size, elevation and SH-201 pressure drop are assumed.	Issue the steam-system P&ID and drum datasheet; re-run this basis with the real drum and SH-201 data.	Relief & vent	OPEN

Key	EQ id	Document	Discipline	Subject	Finding	Proposed resolution	Raised by	Status
L-07	EQ-ep-04	SO3-V1-PE-HMB-001	Process	Rated capacity 110 %	The 110 % case (45 t/h) is outside the operating window 20–44.7 t/h: FV-101 97.1 % open (bound 95 %), PI-200 42.3 kPa g (alarm 42), K-101 4840 kW. The design capacity has only 9.2 % margin to the window top.	Either state the design capacity as 109 % of normal, or rate K-101 head / FV-101 Cv for 110 % flow at PI-200 ≤ 42 kPa g and confirm the converter ΔP at 110 %.	Process engineering	OPEN
L-07	EQ-er-07	SO3-V1-PE-RLB-001	Instrument	PI-200 HI alarm at the top of the normal range	At 45 t/h sulfur (model, loops in AUTO) PI-200 is 42.3 kPa g, above the HI alarm of 42 kPa g; with both air valves wide open it reaches 43.6 kPa g. The alarm would stand during a rate push.	Review the PI-200 HI alarm (e.g. 45 kPa g, still far below the 100 kPa g design) in the alarm rationalisation, or limit the maximum rate.	Relief & vent	OPEN
L-08	EQ-ei-06	SO3-PID-001 rev A	Instrumentation	Position switches for XV-103 and HV-112 not on the P&ID	The start permissive 'HV-112 closed' and the proof of XV-103 closure need position switches; the model reads the valve states directly and neither sheet shows ZSC-112 or ZSC/ZSO-103. The list carries them as assumed (1 DCS DI, 2 SIS DI).	Add ZSC-112 (on the LC valve) and ZSC-103 / ZSO-103 to SO3-PID-001 at rev B and wire ZSC-103 to SIS-100.	Instrumentation & control	OPEN
L-08	EQ-ei-07	SO3-PID-001/002 rev A	Instrumentation	Eight DCS instruments not on the EPC sheets	FI-109, FI-253 and FI-309 are on the reference drawing only; PI-200, TI-207, TI-253, TI-254 and TI-255 are on the DCS but on neither drawing (P&ID tab part 5). They are in this list with P&ID sheet '—'.	Add them to SO3-PID-001/002 at rev B (already an open item of the P&ID tab part 6).	Instrumentation & control	OPEN
L-08	EQ-em-03	SO3-PID-001/002	Process	P&ID rev A design-point table	P&ID table bed ΔP 4.5/3.0/3.6/2.9/3.0 kPa and stack 286 ppm; model PDI-204…244 4.52/2.86/3.16/2.54/2.50 kPa and AIT-306 292.5 ppm.	Update the table at rev B from the model (or the HMB SO3-V1-PE-HMB-001).	Mechanical	OPEN
L-08	EQ-pd-05	SO3-PID-001/002 rev A / SO3-V1-PID-001 rev A	Instrumentation	Instruments on the reference drawing but not on the EPC sheets	FI-109 (H-101 firing), FI-253 (boiler feed water) and FI-309 (acid production, calculated). On the DCS but on neither drawing: PI-200 (converter inlet pressure), TI-207 (superheated steam), TI-253 (E-201 hot gas out), TI-254 (E-202 return gas out) and TI-255 (BFW out of E-203).	Add the eight instruments to both drawings at the next revision (P&ID tab part 6, TBC).	P&ID tab	OPEN
L-08	EQ-pd-06	SO3-PID-001/002 rev A / SO3-V1-PID-001 rev A	Process	Design point table on the sheets	Bed inlet and exit temperatures and conversions agree with the model at design (420→631 °C, 63.1 % · 445→528 °C, 88.1 % · 440→465 °C, 95.8 % · 425→441 °C, 99.70 % · 428→427 °C, 99.80 %). The bed pressure drops on the sheets (4.5 / 3.0 / 3.6 / 2.9 / 3.0 kPa) differ from the model's design values (4.52 / 2.86 / 3.16 / 2.54 / 2.50 kPa), and the sheets quote a stack of 286 ppm SO_2 against 292 ppm in the model.	Re-issue the design-point table on both sheets from the model or the HMB SO3-V1-PE-HMB-001 (P&ID tab part 6, TBC).	P&ID tab	OPEN
L-09	EQ-ep-02	SO3-V1-PE-HMB-001	Process	Heat of sulfur combustion	The model furnace uses K.DHS = 295 kJ/mol (104.6 MW at design); the Process, BFD and Efficiency tabs use 296.8 kJ/mol (105.3 MW).	Adopt one value (the model's 295 kJ/mol in this HMB) and correct the other tabs, or change K.DHS and re-run.	Process engineering	OPEN
L-09	EQ-rs-02	Process / BFD / Efficiency tabs	Process	Heat of sulfur combustion	The model uses K.DHS = 295 kJ/mol; the Process, BFD and Efficiency tabs and the brief use 296.8 kJ/mol (105.3 MW against 104.6 MW).	Adopt one value in all tabs and documents (the HMB keeps the model value).	Project resume	OPEN
L-10	EQ-pd-01	SO3-PID-001/002 rev A / SO3-V1-PID-001 rev A	Process / Document control	Drawing number and revision	The reference drawing's title block reads "SO3-V1-PID-001 rev A"; the EPC set is SO3-PID-001 and SO3-PID-002, also rev A. Two different drawings carry rev A and nearly the same number, and the EPC revision table does not yet name the reference drawing as the revision used in the decks.	Keep SO3-V1-PID-001 for the screen drawing, enter it in the EPC revision table as the revision used by the DCS, decks and exercises (P&ID tab part 6, HOLD), and give it a distinct revision series.	P&ID tab	OPEN
L-10	EQ-ej-01	SO3-V1-PM-DOC-001	Project / Document control	Document numbers outside the scheme	4 of 14 documents do not follow SO3-V1---: PFD-201, SO3-PID-001, SO3-PID-002, SO3-V1-PID-001. They were issued before the scheme (PFD and P&IDs) or carry no discipline code.	Keep the drawing numbers (they are on the title blocks) and add a cross-reference number in this register, e.g. PFD-201 → SO3-V1-PE-PFD-201; SO3-PID-001 → SO3-V1-PE-PID-001; SO3-PID-002 → SO3-V1-PE-PID-002; SO3-V1-PID-001 → SO3-V1-PE-PID-901.	Project & deliverables	OPEN
L-11	EQ-em-05	SO3-V1-ME-EQL-001	Mechanical	R-201 shell thickness: pressure governs in pass 1	At 100 kPa g and 700 °C, 304H (S ≈ 27 MPa, E 0.85) needs 28.5 mm against a handling minimum of 15.5 mm; passes 2–5 need 11.0 mm (minimum governs). "Near-atmospheric, so the minimum governs" holds only below about 631 °C for this shell.	Zone the shell (pass-1 courses at the pass-1 design temperature, lower courses lower); consider internal insulation of pass 1 or a refractory-lined carbon-steel shell; confirm the design pressure basis (1.1 × blower shut-off).	Mechanical	OPEN
L-11	EQ-er-01	SO3-V1-PE-RLB-001	Process	R-201 and the gas train have no relief device	The converter, F-101, B-101 gas side, the exchangers and the towers rely on DP-BASIS: K-101 shut-off is 75.0 kPa g at 90 °C (model) and 81.8 kPa g at 60 °C blower inlet (assumed); the throttled-outlet run settles PI-200 at 74.9 kPa g. The pressure-vessel margins rule (op + 1.7 bar) would give 205 kPa g for R-201, far more than the shut-off needs on a 13 m vessel; the EQL uses 100 kPa g.	Write the gas-train design basis as: design pressure ≥ K-101 shut-off at the coldest blower inlet plus 10 % (≥ 90 kPa g; 100 kPa g assumed) and a vacuum rating for cool-down with the dampers shut (−5 kPa g assumed). Get the certified K-101 curve to confirm the shut-off.	Relief & vent	OPEN
—	EQ-ec-01	SO3-V1-PC-CMS-001	Process / Mechanical	Acid coolers and sulfur filters have no tag or datasheet	PK-08 (acid coolers) and the sulfur filters of PK-09 cannot be requisitioned: PFD-201 does not show them, SO3-PID-002 refers to an acid-system P&ID (note 2) that is not issued, and the filters appear only through PDG-103.	Issue the acid-system P&ID; assign equipment tags (e.g. acid coolers on the IPA and FAT loops, sulfur filters A/B); add datasheets to the equipment list SO3-V1-ME-EQL-001 with the duties from an acid-loop heat balance.	Procurement to commissioning	OPEN

Key	EQ id	Document	Discipline	Subject	Finding	Proposed resolution	Raised by	Status
—	EQ-ec-04	SO3-V1-PC-CMS-001 / Design tab	Mechanical	Converter diameter is an assumption	The converter MR (PK-04) uses inside diameter 13.0 m, which the Design tab labels assumed; bed depths, grid loads and the catalyst loading plan follow from it.	Confirm the diameter in the R-201 datasheet (SO3-V1-ME-EQL-001) before the long-lead MR; record the bed depths for the catalyst loading procedure.	Procurement to commissioning	OPEN
—	EQ-ec-05	SO3-V1-PC-CMS-001 / SO3-V1-IN-SRS-001	Instrumentation / Operations	Proof-test interval versus turnaround interval	The SRS sets a 1-year proof test with a full stroke of XV-103; sulfuric-acid plants typically run 2–3 years between turnarounds, and XV-103 cannot be full-stroked with sulfur flowing without a bypass.	Specify partial-stroke testing or an online test bypass for XV-103 in PK-10, or align the proof-test interval with the turnaround and re-verify the PFD.	Procurement to commissioning	OPEN
—	EQ-ee-02	SO3-V1-EE-LL-001	Electrical	P-301A/B motor voltage	P-301A/B 250 kW at 400 V draws 437 A full load (≈ 2.6 kA starting), above the 300 A LV practice limit used here.	Supply P-301A/B at 690 V (≈ 253 A) or 6.6 kV, or keep 400 V with a soft starter and parallel cables; decide with the MCC design.	Electrical	OPEN
—	EQ-ee-03	SO3-V1-EE-LL-001	Electrical	K-101 starting	Direct-on-line start of the 5000 kW K-101 (6 × FLC, 35.1 MVA) on a 8 MVA, 10 % transformer dips the 6.6 kV bus by about 33.8 % (typical limit 15 %). Soft start (≈ 3 × FLC) 20.3 %; VSD 8.5 %.	Use a VSD on K-101 (also the energy option in the Efficiency tab) or a soft starter; check the upstream fault level (assumed 500 MVA at 22 kV).	Electrical	OPEN
—	EQ-ee-04	SO3-V1-EE-LL-001	Electrical / Process	H_2S basis of the hazardous-area classification	The model has no H_2S. The sulfur-pit classification (Zone 1 inside, Zone 2 around vents, IIB T3) assumes undegassed molten sulfur with dissolved H_2S and a pit sweep-air system that is not on the P&IDs.	Confirm the sulfur specification (degassed or not) and the pit ventilation design; revise the zones and the Ex ratings of P-101A/B accordingly.	Electrical	OPEN
—	EQ-ei-02	SO3-V1-IN-INL-001	Instrumentation	Calibrated ranges not specified	No instrument datasheet exists for this unit. The list takes each range from the DCS display span (src/ui1.js IND / CRANGE), the controller span in makeCtrls(), or the analyzer range stated in measure().	Confirm every calibrated range on the instrument datasheets (Rev A) and re-issue the list; until then the ranges are marked as DCS spans, i.e. assumed.	Instrumentation & control	OPEN
—	EQ-ei-03	SO3-V1-IN-INL-001	Instrumentation / DCS	AIT-306 DCS span narrower than the analyzer range	The stack SO_2 analyzer is ranged 0–5000 ppm (measure()), but the DCS display span of AIT-306 is 0–1500 ppm. The ASHH-306 setting (3000 ppm) is off the top of the DCS bar and trend.	Set the DCS span of AIT-306 to the analyzer range 0–5000 ppm (keep a 0–1500 ppm trend pen for the permit), so the operator sees the approach to the trip.	Instrumentation & control	OPEN
—	EQ-ei-04	SO3-V1-IN-INL-001	Instrumentation	Converter temperature ranges start at 100–300 °C	18 of 25 temperature transmitters (bed inlet, middle and exit TCs, TI-207, TI-253, TI-251) have spans starting at 100–300 °C, but a cold start-up heats the converter from ambient (K.Tamb 30 °C) and the start permissives need TI-201/202 ≥ 400 °C and TI-232 ≥ 360 °C to be seen rising.	Range the transmitters 0–700 °C (beds) and 0–400 °C (TT-251); keep the narrow controller spans in the DCS for tuning only.	Instrumentation & control	OPEN
—	EQ-ei-09	SO3-V1-IN-INL-001	Instrumentation / SIS	Start permissives read DCS measurements	The sulfur admission permissives (TI-106, TI-201, TI-202, TI-232, FI-301, FI-302, HV-112, FIC-103 MAN ≤ 25 %) are checked by the SIS in the model but come from DCS transmitters and a DCS controller state.	Define whether permissives are SIS logic (hardwired or SIS transmitters) or DCS logic releasing an SIS reset; document in the SRS and the control narrative.	Instrumentation & control	OPEN
—	EQ-em-06	SO3-PID-002	Piping / Process	Second-pass line sizes	DN2000-PG-302 40 m/s; DN2000-PG-303 42 m/s; DN2100-PG-304 38 m/s at the design gas flow, against 20–25 m/s in P&ID general note 2.	Resize to about DN2700/DN2800 (22.5 m/s) or confirm the higher pressure drop against the K-101 head.	Mechanical	OPEN
—	EQ-ep-05	SO3-V1-PE-HMB-001	Process / Mechanical	Steam pressure basis	K.Tsat = 250 °C is the saturation temperature at ≈ 3.98 MPa abs, while B-101 steam is specified at 4.0 MPa g (4.10 MPa abs, Tsat 251.8 °C, steam tables).	State steam pressure as gauge on all documents; use 251.8 °C (or set the drum to 3.88 MPa g) on the B-101 datasheet.	Process engineering	OPEN
—	EQ-ep-08	SO3-V1-PE-HMB-001	Process	Converter ΔP basis	PI-200 (35.0 kPa g) is computed from the cold design bed ΔP (ΣK.dPbed = 14 kPa) plus K.dPrest 21 kPa, while PDI-204…244 read the temperature-corrected ΔP (Σ = 15.6 kPa). Their sum plus K.dPrest is 36.6 kPa, not 35.0.	Use one ΔP basis for PI-200 and the PDIs; this HMB keeps the PDIs and allocates the remaining 19.4 kPa to exchangers, towers and ducts (assumed split).	Process engineering	OPEN
—	EQ-er-02	SO3-V1-PE-RLB-001	Mechanical	Steam PSV set against drum design	With the drum design at 4.40 MPa g (EQL / margins rule on 4.0 MPa g) and the PSVs set at it, the drum works at 90.9 % of the PSV set: above the 90 % usually kept so that spring valves do not simmer.	Raise the drum and SH-201 design pressure to ≈ 4.6 MPa g (operating ≈ 87 % of set) or confirm pilot-operated valves; keep PSV-101C on SH-201 set lowest.	Relief & vent	OPEN
—	EQ-er-03	SO3-V1-PE-RLB-001	Process	Blocked-in economizer can steam	With BFW at 10 % of design (model) the E-203 water outlet reaches 462 °C, above Tsat 265 °C, making ≈ 8.7 t/h of steam; a ¾ × 1 in TSV passes 2.23 t/h of steam.	No isolation valve between the economizers and the drum, or car-sealed open; the TSVs cover only liquid expansion during maintenance. Low BFW-flow alarm FI-253 is already in ALM; add the economizer outlet temperature to the steam-system P&ID.	Relief & vent	OPEN
—	EQ-er-05	SO3-V1-PE-RLB-001	Process	EXD-101 burst band	An explosion panel on F-101 must burst above the K-101 shut-off (81.8 kPa g at 60 °C) or a blocked outlet releases SO_2, yet below the shell strength; 85 kPa g (assumed) leaves a narrow band under 100 kPa g.	Decide: rate F-101 for the start-up deflagration and drop EXD-101, or size EXD-101 by NFPA 68 with Pstat in the band and a safe discharge area; keep the H-101 purge (HAZOP R15) either way.	Relief & vent	OPEN
—	EQ-er-06	SO3-V1-PE-RLB-001	Process	Molten sulfur trapped between valves	Steam-jacketed sulfur between XV-103 and FV-103 (and between pump valves) can be blocked in and heated; the Safety-layers register has no thermal relief for it.	Add thermal relief (jacketed, to the sulfur pit) or a procedure that never blocks in a heated section; confirm in the HAZOP close-out.	Relief & vent	OPEN

Key	EQ id	Document	Discipline	Subject	Finding	Proposed resolution	Raised by	Status
—	EQ-pd-02	SO3-PID-001/002 rev A / SO3-V1-PID-001 rev A	Process	One sheet against two	The reference drawing shows the whole converter section on one screen; the EPC set splits it at E-203 / the return-gas lines (sheet 1: air, F-101, B-101, beds 1–3, SH-201, E-201, E-202; sheet 2: E-203, T-301, beds 4–5, E-204, E-205, T-302, stack) with off-sheet connectors.	No change: record the sheet split in the register; the off-sheet connectors are the interface.	P&ID tab	OPEN
—	EQ-pd-03	SO3-PID-001/002 rev A / SO3-V1-PID-001 rev A	Instrumentation	TIC-231	The reference drawing puts the TIC-231 bubble beside SH-201 and TV-231. The EPC sheets draw TIC-231 on sheet 1 at TV-231, with its transmitter TT-231 on sheet 2 at the bed 4 inlet (sheet note 4). The model's loop is the same either way: it measures the bed 4 inlet and moves TV-231.	No change to the loop; add the TT-231 / TIC-231 split to the instrument list note (it already is: SO3-V1-IN-INL-001).	P&ID tab	OPEN
—	EQ-pd-07	SO3-PID-001/002 rev A / SO3-V1-PID-001 rev A	Instrumentation / SIS	ESD pushbutton tag	The EPC interlock table names it HS-001; the simulator's SIS screen and the reference drawing call it the ESD pushbutton (trip I-6) without a tag number.	Tag the ESD push-button HS-001 on the SIS screen and the screen P&ID, as on the EPC sheets.	P&ID tab	OPEN
—	EQ-pd-08	SO3-PID-001/002 rev A / SO3-V1-PID-001 rev A	Mechanical	Equipment shown only on the EPC sheets	acid pump tanks TK-301/302, mist eliminators, the stack ST-301 as a vessel, the H-101 burner package boundary, line numbers and flow elements. None of these is modelled except through the existing tags.	Keep them on the EPC sheets and the equipment list; note in the model scope that they are not simulated.	P&ID tab	OPEN
—	EQ-rs-03	src/model.js	Process (simulator model)	Stack SO_2 after a sulfur stop	In the model the stack SO_2 drops to zero as soon as sulfur stops: the gas holdup of beds 4–5 and the towers is not modelled, so a trip looks cleaner at the stack than a plant would.	Add a first-order holdup (residence time from the gas volume of R-201 passes 4–5 and T-302) or state the limitation where stack values after a trip are used.	Project resume	OPEN
—	EQ-rs-04	Operating windows tab NOTES.md	Process	Monte Carlo result in the notes	The Operating-windows notes quote 78.3 % for the Monte Carlo result; the function owMC gives 83.7 % (the decks use 83.7 %).	Correct tabs/opwin/NOTES.md to the value the page computes.	Project resume	OPEN
—	EQ-rs-05	Efficiency tab NOTES.md	Process / Electrical	Blower specific energy at 45 t/h	The Efficiency notes give 32.6 kWh/t for constant speed at 45 t/h; the page computes 31.9 kWh/t.	Correct tabs/effic/NOTES.md to the computed value.	Project resume	OPEN
—	EQ-ej-02	SO3-V1-PM-DOC-001	Project / Document control	Documents held only in a tab	SO3-V1-PID-001 (P&ID tab), SO3-V1-IN-SRS-001 (SRS of SIF tab) exist only as page content: there is no issued PDF, so the revision cannot be frozen or transmitted.	Issue each as a PDF from the data its tab already reads (sf.json for the SRS; the #pidSvg drawing for the screen P&ID) and add it to the published files.	Project & deliverables	OPEN
—	EQ-ej-03	SO3-V1-PM-DOC-001	Project / Document control	Two revision conventions in one set	Letter revisions (SO3-PID-001 rev A, SO3-PID-002 rev A, SO3-V1-PID-001 rev A, SO3-V1-SA-HZP-001 rev A) sit beside number revisions (10 documents at rev 0). In typical practice letters mean before IFC (IFR/IFA) and numbers from IFC on, so a reader would take the P&IDs as less mature than the lists drawn from them.	Adopt one convention for the teaching set (e.g. all at rev A "Issued for review" until a checker signs, then rev 0 IFC) and re-issue the covers.	Project & deliverables	OPEN
—	EQ-ej-04	SO3-PID-001 / SO3-PID-002 / SO3-V1-PE-RLB-001 / SO3-V1-IN-INL-001 / SO3-V1-EE-LL-001 / SO3-V1-IN-CE-001	Project / Document control	Input documents not cited	13 of 36 interfaces in part 4 are not cited by number in the receiving PDF: SO3-PID-001 does not cite PFD-201; SO3-PID-002 does not cite PFD-201; SO3-V1-PE-RLB-001 does not cite SO3-V1-PE-HMB-001, SO3-PID-001, SO3-PID-002, SO3-V1-SA-HZP-001; SO3-V1-IN-INL-001 does not cite SO3-V1-PE-HMB-001, SO3-V1-PID-001; SO3-V1-EE-LL-001 does not cite SO3-V1-PE-HMB-001, SO3-V1-IN-INL-001; SO3-V1-IN-CE-001 does not cite SO3-PID-001, SO3-PID-002, SO3-V1-SA-HZP-001.	Give every document a "References" list with the number and revision of each input, so a revision of the input flags the documents to re-check (typical document-control practice).	Project & deliverables	OPEN
—	EQ-ej-05	SO3-V1-PM-DOC-001	Project / Document control	Documents named but not issued	The issued documents refer to deliverables that do not exist in this set: the acid-system P&ID (SO3-PID-002 note 2, EQ-ec-01), the steam-system P&ID, drum datasheet and utility HMB (EQ-ep-07, EQ-er-08, EQ-ec-03), instrument datasheets and loop diagrams (EQ-ei-02, SO3-V1-IN-INL-001 part 1), and a plot plan / layout (Layout & isometrics, step 1.6).	Add them to the register as "planned" with a number reserved (e.g. SO3-V1-PE-PID-003 acid system, SO3-V1-PE-PID-004 steam system, SO3-V1-IN-IDS-001 instrument datasheets, SO3-V1-ME-PLT-001 plot plan) so the gaps are tracked.	Project & deliverables	OPEN

7 · Checks against the model and the other documents

Computed by the same function the tab runs (ejChecks) in node at generation; the tab recomputes them and adds a live check that each document's tab is in the page edition.

#	Check	Found	Result
C1	Every document in the register has its PDF file present	12 of 14 have a PDF; no PDF: SO3-V1-PID-001, SO3-V1-IN-SRS-001	✗
C2	Every EPC tab holds at least one document	7 EPC tabs, each with ≥ 1 document	✓
C3	Every query has a discipline and a proposed resolution	58 queries, all complete	✓
C4	Document numbers follow SO3-V1---	10 of 14 follow it; outside: PFD-201, SO3-PID-001, SO3-PID-002, SO3-V1-PID-001	✗
C5	No duplicate document numbers	14 numbers, all unique	✓
C6	No duplicate query ids	58 ids, all unique	✓
C7	Every linked key groups ≥ 2 existing queries, none in two groups	11 keys, 30 queries linked	✓
C8	Every document number a query names is in the register	all numbers named in the query "doc" fields are registered	✓
C9	Every interface in part 4 joins two registered documents	37 interfaces	✓
C10	Every document (except this register) has at least one input or output	none isolated	✓
C11	Every input of a document is cited by number in its PDF	23 of 36 cited; not cited: PFD-201 in SO3-PID-001, PFD-201 in SO3-PID-002, SO3-V1-PE-HMB-001 in SO3-V1-PE-RLB-001, SO3-V1-PE-HMB-001 in SO3-V1-IN-INL-001, SO3-V1-PE-HMB-001 in SO3-V1-EE-LL-001, SO3-PID-001 in SO3-V1-PE-RLB-001, SO3-PID-002 in SO3-V1-PE-RLB-001, SO3-PID-001 in SO3-V1-IN-CE-001, SO3-PID-002 in SO3-V1-IN-CE-001, SO3-V1-PID-001 in SO3-V1-IN-INL-001, SO3-V1-SA-HZP-001 in SO3-V1-PE-RLB-001, SO3-V1-SA-HZP-001 in SO3-V1-IN-CE-001, SO3-V1-IN-INL-001 in SO3-V1-EE-LL-001	✗
C12	Query register total equals the sum of its sources	58 in the register; sources ecomm 5 + eelec 4 + einst 9 + emech 6 + eproc 8 + erel 8 + pid 8 + RESUME 5 + eproj 5 = 58	✓
C13	Every query carries status OPEN (none closed in a teaching set)	all OPEN	✓

8 · Assumptions

- The EPC phases, revision codes and rules of credit are typical practice, assumed for teaching; no contractor procedure is reproduced.
- Discipline of the legacy drawings (PFD-201, SO3-PID-001/002, SO3-V1-PID-001) taken as Process (PE).
- The interfaces in section 5 are design intent written for this register; whether they are cited is read from the PDF text.
- Page count is used as the progress weight; documents held only in a tab weigh 2.
- Linked keys are an engineering judgement of which queries say the same thing; every original id is kept.
- No client, contractor or vendor is named or represented.